Nguyen Manh Hiep

Software Engineer

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CAREER OBJECTIVE

AI-native full-stack software engineer: I bring LLMs and automation into products, and use AI agents as everyday development tools.
Short term (1–2 years): Join a product team to build and run production AI features (LLM integration, agents, RAG) and the backend behind them, while raising my own standards in testing, code review and system monitoring.
Long term (3–5 years): Grow into a Senior Software Engineer specialising in AI systems, owning a system's technical direction from design to operation and helping the team use AI safely in its development workflow.

EXPERIENCE

Notarex - notarial document digitization platform 03/2026 - 09/2026

Software Engineer, ChimAI JSC (Full-stack + AI document processing + Infrastructure):

- **Description:** A platform that digitizes notarial paperwork for notary offices in Vietnam, covering configurable form templates, AI-driven OCR extraction and a full document edit, save and download cycle.
- **Technologies & Tools:** Next.js 14, TypeScript, FastAPI, SQLAlchemy async, PostgreSQL 16, Celery, Redis, MinIO, OnlyOffice
- **Key Responsibilities & Contributions:**
  - **AI OCR Pipeline:** Generated each document type's extraction prompt from its template configuration, so new document types are added through configuration alone; national IDs are read from their QR code and skip the model.
  - **Document Workflow & Authorization:** Built the flow from uploaded photos through human review to a JWT-protected OnlyOffice edit-save-download cycle with docxtpl DOCX generation, under a three-role model scoped per notary office.
  - **Background Processing:** Built a Celery and Redis task queue on a ZSET pending queue with an atomic active-counter, plus heartbeat reconciliation that detects and reruns zombie jobs so no record gets stuck mid-OCR.
  - **Requirements & Delivery:** Collected requirements directly with notary offices, scoped each release in sprints in a 2 to 3 person team with no dedicated PM, and shipped on Docker behind Nginx with MinIO storage and GitHub Actions CI/CD.

EazySale - sales lead generation and outreach SaaS 03/2026 - 08/2026

Software Engineer, ChimAI JSC (Backend + React dashboards + Infrastructure):

- **Description:** A SaaS for sales lead management and automated customer outreach over Zalo and email. I owned the FastAPI backend, three React dashboards and the infrastructure.
- **Technologies & Tools:** React 18, TypeScript, FastAPI, SQLAlchemy, Alembic, PostgreSQL 15, APScheduler, Zalo OA API, SendGrid, Docker
- **Key Responsibilities & Contributions:**
  - **Zalo OA Integration:** Built the Zalo OA ZNS API channel with template management, HMAC-SHA256 webhook verification and delivery metrics, plus a mock provider with a tunable failure rate for testing without spending quota.
  - **Scheduled Campaign Delivery:** Paired APScheduler date triggers with a 60-second database poll that recovers jobs lost to a restart, using SELECT FOR UPDATE NOWAIT so each message is sent exactly once.
  - **Email Event Tracking:** Made the public SendGrid event webhook idempotent with in-batch deduplication and a status-priority guard inside the UPDATE, so out-of-order and repeated events cannot roll a delivery state backwards.
  - **Architecture & Delivery:** Designed a path-routed Nginx deployment on one domain, RBAC across 129 REST endpoints, and a GitHub Actions pipeline deploying four images with migrations and zero downtime.

ASC Farm - farm production management system 06/2026 - 08/2026

Freelance Software Engineer, sole developer (Backend + Web admin + Mobile PWA):

- **Description:** A web admin and offline-capable mobile PWA that runs an intercropped farm from cultivation plans to verified field tasks.
- **Technologies & Tools:** Frappe v15 (headless), Python, MariaDB 11, Redis, React 18, TypeScript, Leaflet, PWA, Docker
- **Key Responsibilities & Contributions:**

- **Task Generation Engine:** Generated field tasks idempotently from cultivation processes, with prerequisite chains and "N times every N days" frequencies, and regenerated them when a process or crop cycle changes.
- **Workload Balancing:** Assigned each task to a team lead, preferring the block owner and otherwise the least-loaded lead, and reassigned open work automatically when a lead became inactive.
- **Offline-first Field App:** Built an idempotent offline queue keyed by a client UUID, a service-worker app shell, and in-app completion photos stamped with GPS and a watermark as proof of work.
- **Testing & Delivery:** Covered the engine, status model and permissions with 139 backend tests across 65 API endpoints, packaged as a one-command Docker Compose stack.

PROJECTS

SenAI - real-time multilingual meeting translation 2026 - present

Frontend, Backend and AI Integration Owner (Real-time systems + LLM orchestration + Infrastructure):

- **Description:** Live meeting translation by a 5-person team for VAIC 2026 (three awards), running at app.senai.online.
- **Technologies & Tools:** Next.js 16, TypeScript, FastAPI, PostgreSQL + pgvector, LiveKit, vLLM, Soniox, ElevenLabs
- **Key Responsibilities & Contributions:**
  - **AI/LLM Integration:** Built a model router for ASR, translation, TTS and LLM that switches provider at runtime between self-hosted vLLM and commercial APIs, falling back to the self-hosted model when a key is missing or a provider fails.
  - **Translation Pipeline:** Ran an independent ASR and translation pipeline per microphone to separate speakers, pairing fast translation with LLM refinement against a contextual glossary.
  - **Meeting Intelligence:** Added Soniox transcription that separates three or more speakers, LLM-written minutes with an assistant that answers questions about them, and per-speaker voiced translation with ElevenLabs.
  - **Delivery & Security:** Deployed through GitHub Actions over SSH with a forced command that skips API restarts for doc-only commits, revoked refresh tokens on logout, and kept 37 UAT suites with 147 real, non-mocked tests.

Ely - personal AI assistant built on OpenClaw 2026

Extended the open-source OpenClaw agent with Telegram commands, CDP browser automation, cron jobs for daily chores, my own skills that dispatch and supervise Claude Code tasks, and a self-restarting gateway.

EDUCATION

Hanoi University of Science and Technology (HUST) 2021 - 2026

Computer Science

HONORS & AWARDS

Third Prize (top 3 overall of 300+ teams), Vietnam AI Innovation Challenge 2026
Vietnam AI Stars Award, Vietnam AI Innovation Challenge 2026
AI Singapore Award, Vietnam AI Innovation Challenge 2026

TECHNICAL SKILLS

AI-assisted development	Claude Code (agents, skills, hooks), OpenClaw agents, MCP servers, Cursor, Copilot, prompt & context engineering, RAG, LLM orchestration, model routing with fallback, vLLM
Frontend	React 19/18, Next.js 16/14, TypeScript, Vite, Tailwind v4, Zustand, PWA / offline-first
Backend	FastAPI, Python, Frappe v15, REST API design, Node.js / NestJS, Spring Boot, WebSocket, LiveKit, Celery, Redis queues, JWT / OAuth, RBAC, webhooks
Databases	PostgreSQL (+ pgvector), MariaDB, MySQL, SQLite, SQLAlchemy 2.0 async, MinIO (S3)
DevOps & Cloud	Docker Compose, Nginx, systemd, TLS, VPS deployment, GitHub Actions CI/CD
Engineering practice	System design, SOLID, design patterns, code review, idempotency & concurrency control, workflow & scheduling engines, non-mocked integration testing, SRS / Use Case / ERD
Soft Skills	Critical thinking, technical research, cross-functional teamwork, fast adaptability